$$\frac{1}{f} = \frac{1}{30} - \frac{1}{(-10)}$$

$$\frac{1}{f} \Rightarrow \frac{1+3}{30} = \frac{4}{30}$$

$$f = 7.5 \text{ cm}$$

The focal length of the lens is 7.5 cm

10. Myopia can be corrected by using concave lens. The lens must be placed that an object at infinity forms its image on far point.

Hence, $f = -80$ cm
$f = -0.8$ m

$$P = \frac{1}{f}$$

$$P = \frac{1}{-0.8\text{m}} \Rightarrow -1.25\text{ D}$$

11. (a) Convex lens

(b) The image is real and inverted in nature hence $\frac{v}{u}$ (magnification) is negative.